HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and fine-tuned **Large Language Models (LLMs)** for automated red-teaming and vulnerability detection, achieving a **15% reduction** in simulated attack success rates.
- Engineered and deployed **AI/LLM-powered features** into production security platforms, optimizing for **inference speed** and **resource utilization**.
- Built and maintained **MLOps pipelines** for end-to-end AI/LLM workflows, including data preprocessing, fine-tuning, and deployment monitoring.
- Collaborated with AI researchers and software engineers to productionize **state-of-the-art AI/LLM techniques** for security applications.
- Implemented strategies to detect and mitigate **LLM security vulnerabilities** such as prompt injection and data poisoning.
- Containerized AI/LLM model inference services using **Docker** and deployed them to a **Kubernetes cluster** on AWS, managing configurations with Helm.
- Designed and implemented Python scripts to automate data ingestion and preprocessing for ML model training, handling approximately **2,000 records per minute**.

SKILLS

AI/ML – PyTorch, TensorFlow, Hugging Face Transformers, LLMs, Intelligent Agents, Deep Learning, RLHF, AI Security, Prompt Injection, Data Poisoning

MLOps – Model Deployment, Pipeline Development, Validation, Monitoring, Inference Optimization, Cost-Efficiency, Resource Utilization

Programming – Python, Java, SQL, Bash Scripting

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Helm

Data Tools – pandas, NumPy, scikit-learn, Vector Databases, PostgreSQL, MySQL, Oracle

PROJECTS

AI-Powered Vulnerability Detection System

- Developed a Python-based system to detect vulnerabilities in AI models using **fine-tuned transformers** and **adversarial attack simulation** techniques.
- Implemented **Reinforcement Learning from Human Feedback (RLHF)** to align model outputs with security policies, improving detection accuracy by **18%**.
- Containerized the detection engine with **Docker** and deployed it on **AWS Lambda** for scalable, on-demand analysis of security logs.

Secure LLM Inference Service

- Built a **FastAPI** service in Python to host a fine-tuned LLM for generating security-related content, achieving p95 latency below **200ms**.
- Integrated **input validation and output sanitization** to mitigate prompt injection attacks, processing approximately **500 requests per hour**.
- Configured the service for deployment on a **Kubernetes cluster**, including resource limits and auto-scaling policies.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)